KULLIYYAH OF ENGINEERING

KAGGLE COMPETITION

LLM AGENTIC LEGAL INFORMATION RETRIEVAL

MCTA 4363 DEEP LEARNING

ASSESSMENT 2

INSTRUCTOR: DR HASAN FIRDAUS BIN MOD ZAKI

NAME	Matric No.
MUHAMMAD EIQBAL BIN HASBOLLAH	2216911
IKMAL FIKRI BIN KHAIRUL KHUBAIDILLAH	2218723
MUHAMMAD ZAMIR FIKRI BIN MOHD ZAMRI	2212515

Section 1

SEM 2, 2025/2026

1. Overview

Assessment 2 replaces every component of the baseline with a state-of-the-art hybrid retrieval pipeline. It introduces lexical BM25 search alongside dense FAISS retrieval, fuses them with **asymmetrically weighted Reciprocal Rank Fusion**, extends coverage to the full court decisions corpus via a three-variant multi-query strategy, and applies a **Qwen3 generative neural reranker with chunk-level MaxSim scoring** to re-order candidates before final output.

2. Model Stack

Role	Model / Library	Purpose
Dense embedder	Qwen/Qwen3-Embedding-0.6B	High-quality multilingual embeddings with instruction prefix; fp16 inference
Neural reranker	Qwen/Qwen3-Reranker-0.6B	Generative yes/no relevance scoring over (query, document) pairs; uses LM logits
Sparse index	BM25s (bm25s library)	Fast lexical retrieval for German law abbreviations and exact article keywords
Dense index	FAISS IndexFlatIP	Inner-product nearest-neighbour search over law corpus embeddings; disk-cached

3. Full Pipeline (Step by Step)

Step 1 — Laws Hybrid Retrieval

BM25 and FAISS each independently retrieve the top-60 law candidates for the query. Their ranked lists are merged using **asymmetrically weighted RRF**: FAISS rank receives weight ×1.5 (dense semantic signal prioritised); BM25 rank receives weight ×0.5 (lexical signal used as supplementary signal). The combined score is

score(d) = 0.5 / (60 + rank_BM25) + 1.5 / (60 + rank_FAISS)

This weighting reflects the observation that semantic embedding provides a stronger cross-lingual signal than lexical matching for English-to-German queries, while still benefiting from BM25 for abbreviation-heavy queries.

Step 2 — Court Decisions Multi-Query Retrieval

Because court decisions are only indexed via BM25 (no FAISS index for the large court corpus), three distinct queries are issued independently to maximise recall:

- Query A — Original English query: captures direct semantic match from English terms the model recognises.
- Query B — Query + top-3 retrieved law citation strings: appending citation text as pseudo-document expansion bridges the language gap using already-retrieved German keywords.
- Query C — Law abbreviations only (StPO, OR, ZGB, BV, etc.): uppercase tokens extracted from top-10 retrieved law citations act as high-precision legal domain filters for court decision retrieval.

Results from all three queries are merged by **max-pooling BM25 scores;** each court document retains the highest score it received across any query variant.

Step 3 — Candidate Merging & Deduplication

Law citations (from hybrid retrieval) and court citations (from multi-query BM25) are concatenated, deduplicated, and capped at 60 total candidates for the reranking stage.

Step 4 — Chunking + MaxSim Reranking

Each candidate document is split into **overlapping 200-word chunks (50-word overlap)**. Every chunk is independently scored by the Qwen3-Reranker using a yes/no generative prompt. The **maximum chunk score (MaxSim)** is assigned to the parent document. This ensures relevant passages buried deep in long court decisions are surfaced rather than lost to input truncation.

Step 5 — Adaptive Citation Count

All citations with a reranker score **≥ 0.5** are included unconditionally. A minimum of 10 citations is always returned (for low-confidence queries), with a hard cap of 30 (to control false positives). This dynamically matches output volume to query complexity.

Step 6 — Disk Caching

Both BM25 indexes (laws and courts) and the FAISS laws index are serialised to disk after first construction. Subsequent runs within the same Kaggle session load from cache, avoiding the cost of re-encoding the full corpus.

4. Hyperparameters

Parameter	Value	Purpose
TOP_K_RETRIEVAL	60	Candidate pool size from BM25 and FAISS before fusion
TOP_K_FINAL (min)	10	Minimum citations returned per query regardless of threshold
RRF_K	60	RRF smoothing constant; balances rank sensitivity
FAISS weight (RRF)	1.5	Upweights dense retrieval signal in fusion
BM25 weight (RRF)	0.5	Downweights sparse retrieval to supplementary role
CHUNK_SIZE	200 words	Length of each text chunk for MaxSim reranking
CHUNK_OVERLAP	50 words	Overlap between adjacent chunks to avoid boundary cutoff
TEXT_TRUNCATE	3,500 chars	Maximum text fed to reranker per chunk
LAWS_EXPANSION_TOP_K	3	Law citations appended to court Query B expansion
Reranker threshold	0.5	Minimum yes-probability for automatic citation inclusion
Max citations output	30	Hard cap on predictions per query
EMBEDDING_BATCH_SIZE	16	Batch size for corpus encoding (GPU memory tuned)
RERANKER_BATCH_SIZE	8	Batch size for reranker inference

max_seq_length	512 tokens	Query encoder max sequence length

5. Assessment 1 (Baseline) vs. Assessment 2 — Direct Comparison

Aspect	Assessment 1	Assessment 2	Δ
Kaggle Macro F1	0.01250	**0.07013**	**+5.8 %**
Avg citations/query	20 (fixed)	18.1 (adaptive)	↑
Embedding model	MiniLM-L12 (117M)	Qwen3-Embedding-0.6B (600M)	↑↑
Reranker	None	Qwen3-Reranker-0.6B (yes/no gen.)	**New**
Retrieval	Dense only (cosine)	BM25 + FAISS + weighted RRF	↑↑
RRF weighting	N/A	FAISS ×1.5 / BM25 ×0.5 (asymmetric)	**New**
Court decisions	Not used	Multi-query BM25 (3 variants)	**New**
BM25 disk cache	N/A	Both laws & courts cached	**New**
FAISS disk cache	N/A	Laws FAISS index cached	**New**
Document chunking	None (full truncation)	200-word chunks + MaxSim	**New**

Output count	Fixed top-5	Adaptive: threshold + min/max bounds	↑
Task instruction	Generic multilingual retrieval	English-to-German legal specific	↑
Citation quality	Noisy (garbled strings observed)	**Clean canonical citations**	↑↑

6. Iterative Development Log

Property	Value
Embedding model	Qwen/Qwen3-Embedding-0.6B (fp16, max_seq=512)
Reranker	Qwen/Qwen3-Reranker-0.6B (generative yes/no)
Retrieval corpus	laws_de.csv + court_considerations.csv
Laws retrieval	BM25 + FAISS with asymmetric RRF (FAISS ×1.5, BM25 ×0.5)
Court retrieval	Multi-query BM25: original + law expansion + abbreviations
Index caching	BM25 (laws + courts) and FAISS serialised to disk
Task instruction	Given a legal question in English, retrieve relevant Swiss law articles and court decisions written in German.
Chunking	200-word overlapping chunks (50-word overlap), MaxSim aggregation
Output strategy	Adaptive: all scores ≥ 0.5, minimum 10, maximum 30 citations
Validation F1	0.0700
Kaggle LB F1	0.0700 (+5.9% relative vs. baseline)
Key observation	Citation quality improved dramatically — submission now contains clean canonical strings (Art. 59 Abs. 1 SVG, Art. 166 Abs. 2 SchKG). Asymmetric RRF weighting proved important: equal weights reduced score compared to FAISS-prioritised fusion. Adaptive output count better matches gold citation volumes (avg 18.1 vs fixed 20). Main remaining

	bottleneck: no FAISS dense index for the court corpus; court recall relies solely on BM25.

7. Techniques Tried & Insights Gained

7.1 Multilingual Dense Embeddings with Instruction Prefix

Technique: Replacing MiniLM with Qwen3-Embedding-0.6B and prepending a domain-specific instruction string before encoding queries.

The instruction prefix *"Instruct: Given a legal question in English, retrieve relevant Swiss law articles and court decisions written in German. Query: <q>"* activates the model's retrieval-optimised representation pathway. Without it, Qwen3-Embedding falls back to general representation mode and underperforms MiniLM on retrieval benchmarks.

Insight: The specificity of the instruction matters. Mentioning 'English' query language and 'German' document language explicitly improves cross-lingual alignment. This is the most impactful single change in the upgrade.

7.2 Asymmetrically Weighted Reciprocal Rank Fusion

Technique: Combining BM25 and FAISS rankings with weights 0.5 and 1.5 respectively (rather than equal weights of 1.0 each).

Standard RRF (equal weights) treats both retrievers as equally reliable. However, for cross-lingual queries (English → German), the dense semantic model provides a much stronger signal than lexical matching. Upweighting FAISS captures this asymmetry and avoids BM25 noise from English-German vocabulary mismatch dragging relevant documents down the ranking.

Insight: Weight tuning is a low-cost hyperparameter that significantly affects fusion quality. The 3:1 ratio (FAISS:BM25) worked well here. However, BM25 still contributes important recall for queries containing exact legal abbreviations (StPO, OR, ZGB) that the embedding model may not align well cross-lingually.

7.3 Three-Variant Multi-Query Court Retrieval

Technique: Issuing three independent BM25 queries against the court decisions corpus and merging by max-score.

- Query A (original): The English query directly. Captures any English tokens that exist in German court texts and provides a baseline.
- Query B (law expansion): The English query concatenated with the top-3 retrieved law citation strings (e.g. "Art. 59 Abs. 1 SVG"). These strings are already in German and act as a vocabulary bridge, improving cross-lingual BM25 recall.
- Query C (abbreviations only): Uppercase tokens extracted from the top-10 retrieved law citations (e.g. "SVG ZPO SchKG"). These high-precision legal domain markers directly target the relevant area of Swiss law in court decisions.

Insight: Query C (abbreviations) is the most cost-effective expansion strategy because law abbreviations are language-agnostic and appear verbatim in both laws and court decisions. It effectively performs implicit query translation without any translation model.

7.4 Chunking with MaxSim Reranking

Technique: Splitting candidate documents into 200-word overlapping chunks and assigning each document its maximum chunk-level reranker score.

Court decisions can span thousands of words. Truncating to 3,500 characters (the reranker input limit) discards most of the document. A relevant holding or citation may appear late in the decision. Chunking ensures the reranker evaluates every passage, and MaxSim (taking the highest-scoring chunk per document) assigns credit wherever relevance is strongest.

Insight: The 200-word chunk size with 50-word overlap balances granularity against the number of reranker calls (which grows linearly with chunk count). For the 60-candidate reranking pool, the total number of chunk-pairs is manageable within the 12-hour compute constraint. Smaller chunks would increase recall at the cost of more inference time.

7.5 Qwen3 Generative Yes/No Reranking

Technique: Using Qwen3-Reranker-0.6B to score (query, document) relevance by extracting the softmax probability of the token 'yes' at the final sequence position.

The model is prompted with a structured template: System (judge relevance), User (Instruct, Query, Document), then the assistant generates with the reasoning chain suppressed (*<think>\n\n</think>*). The logit over 'yes' vs 'no' tokens at the last position is treated as a calibrated relevance score.

Insight: Generative rerankers leverage the full pre-trained knowledge of an LLM rather than a narrow classification head. They tend to outperform cross-encoder classifiers on multilingual and low-resource tasks. The yes/no framing is also robust to domain shift — the model can reason about whether a German law article satisfies an English legal question without explicit translation supervision.

7.6 Adaptive Citation Count via Confidence Threshold

Technique: Returning all citations where the reranker confidence exceeds 0.5, subject to a minimum of 10 and a maximum of 30.

Gold citation counts per query vary widely (ranging from 2 to 15+ in the training set). A fixed output count of 5 guarantees poor recall for multi-citation queries; a fixed count of 20 guarantees poor precision for simple queries. Anchoring the output to reranker confidence dynamically adapts to query complexity.

Insight: The threshold value (0.5) was set based on the Qwen3-Reranker output distribution where scores tend to be bimodal (clearly relevant vs. clearly irrelevant). The minimum of 10 is a safety floor for queries where the reranker assigns low scores across the board. Both parameters are tunable and represent an optimisation opportunity.

7.7 Index Disk Caching

Technique: Serialising BM25 indexes (both laws and courts) and the FAISS laws index to disk after first construction.

Insight: For Kaggle environments with session persistence, caching avoids re-encoding the entire corpus on every run. For the court corpus (which is very large), BM25 construction is the most expensive step. Caching both indexes within the same session is critical for iterative experimentation within the 12-hour compute window.

8. Next Steps to Increase the Score

8.1 Build a FAISS Dense Index for Court Decisions (Highest Priority)

Currently, the court corpus uses BM25 only. The single largest recall gap is the absence of dense retrieval for court decisions. Adding an FAISS index over Qwen3-Embedding representations of the court corpus, and applying the same BM25 + FAISS + asymmetric RRF pipeline used for laws, is the most direct next improvement.

- Encode court_considerations.csv with Qwen3-Embedding in batches (to manage GPU memory for the large corpus).
- Build FAISS IndexFlatIP and serialise to disk — essential given corpus size.
- Apply the same FAISS ×1.5 / BM25 ×0.5 weighted RRF as used for laws.

Expected impact: High — court recall is the primary bottleneck. Dense search would surface semantically relevant decisions that share no English vocabulary with the query.

8.2 Fine-Tune the Embedding Model on Training Data

Both models are general-purpose. Fine-tuning Qwen3-Embedding on the competition's train.csv with contrastive learning would specialise it for Swiss legal domain retrieval.

- Create positive pairs: (query, gold_citation_text) from train.csv.
- Mine hard negatives: top-ranked but non-gold citations from the current retrieval pipeline make the most informative negatives.
- Train with MultipleNegativesRankingLoss (suitable for single positive per query).

Expected impact: High — domain fine-tuning typically yields 5–15% relative improvement on specialised retrieval tasks. The training set provides direct supervision signal.

8.3 Tune the Adaptive Threshold on the Validation Set

The reranker threshold (0.5), minimum citations (10), and maximum citations (30) were chosen heuristically. Grid search over these three parameters using val.csv's 10 queries with known gold citations could directly optimise the precision-recall trade-off.

- Grid search: threshold $\in$ {0.3, 0.4, 0.45, 0.5, 0.55, 0.6}

- Grid search: min_citations ∈ {3, 5, 7, 10}
- Grid search: max_citations ∈ {15, 20, 25, 30}
- Evaluate each combination with macro F1 on val.csv and select the globally optimal triplet.

Expected impact: Medium — the current values are reasonable but not empirically validated. Even a 0.05 threshold shift can move the F1 by 1–3 points relative.

8.4 German Query Translation for BM25 Boost

BM25 on court decisions receives only English queries (and query expansions derived from already-retrieved law citations). Explicitly translating queries to German would enable BM25 to directly match German legal vocabulary in court decisions.

- Use a small multilingual LLM (e.g. Qwen2.5-1.5B) to translate each test query to German.
- Issue the German translation as Query D in the court multi-query strategy, merged by max-pool.
- Alternatively: translate queries and run BM25 on the German translation only, replacing Query A.

Expected impact: Medium — particularly beneficial for queries with no German-adjacent vocabulary in the English original.

8.5 Larger or Ensemble Reranker

Qwen3-Reranker-0.6B is compact. Larger variants (1.7B, 4B) or alternative cross-lingual rerankers (e.g. BGE-Reranker-v2-m3) may provide stronger relevance discrimination.

- Test Qwen3-Reranker-1.7B if GPU memory allows (within 12-hour budget).
- Test BGE-Reranker-v2-m3 as an alternative multilingual cross-encoder.
- Ensemble two rerankers by averaging scores — tends to reduce individual model errors.

Expected impact: Medium — larger models improve precision at the top of the reranked list. Most beneficial when the retrieval recall is already high.